BELA Biodiversity, Ecosystems, & Landscape Assessment | PROGREEN | THE WORLD BANK IBRD • IDA | WORLD BANK GROUP

BELA CASE STUDY SERIES: ETHIOPIA

DEVELOPING AN INVESTMENT PRIORITIZATION TOOL TO COMBAT LAND DEGRADATION

CASE STUDY PROFILE

WORLD BANK REGION

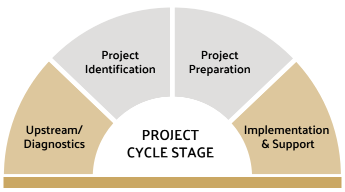
Sub-Saharan Africa

SCALE OF ANALYSIS: National & Project area

BANK PROCESS INFORMED

- Advisory Services and Analytics (ASA)
- Country Climate and Development Report (CCDR)

PROJECT CYCLE STAGE



SECTORS ENGAGED

- Agriculture
- Environment & Natural Resources
- Finance
- Climate Change
- Poverty

ECOSYSTEM SERVICES

- Erosion & Sedimentation
- Carbon Storage
- Water Regulation
- Flood Mitigation
- Crop Production
- Livestock Production

INSIGHTS

- Land condition & trends
- Landscape investment priority areas
- Ecosystem services in macro economy
- Climate change impacts
- Carbon Market Potential
- Support For Natural Capital Accounting

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BACKGROUND

Situated in the Horn of Africa, Ethiopia spans 1.1 million km^2. It is the second most populous country in Sub-Saharan Africa, with a population of around 120 million, 80 percent of whom live in rural areas. The country has made significant strides in economic and human development, with annual growth averaging 11 percent since 2004 and a remarkable reduction in extreme poverty from 55 percent in 2000 to 26.7 percent in 2016. Despite these achievements, the country grapples with interlinked challenges such as environmental sustainability, economic resilience, and climate vulnerability, reflected in its ranking of 163 on the 2021 ND-GAIN Index.

Ethiopia confronts a pressing environmental crisis– land degradation – that significantly hinders agricultural productivity. Characterized by soil erosion and nutrient depletion, this problem was particularly acute in the highlands in the 1980s, stemming from factors such as weak land tenure arrangements, population growth, intensive land use, and inadequate. Farming practices such as free grazing, tree removal, and inappropriate tillage exacerbate soil impact, escalating vulnerability to food and water insecurity. Confronting these challenges necessitates a comprehensive strategy, harnessing sustainable land management practices (SLM) and strengthening land tenure to concurrently address climate change, improve agricultural productivity, safeguard biodiversity, and enhance human well-being.

The World Bank is assisting the Government of Ethiopia (GoE) to tackle these challenges through investments in sustainable land management; the Government has also expressed interest in shifting these investment programs away from being input-based to rewarding communities for performance measured by improvements in ecosystem services provision. The Country Climate and Development Report (CCDR) for Ethiopia also provided an additional entry point to understand the interplay between climate change and land degradation, and to identify policy measures to support climate resilience and low carbon development. For the CCDR itself, the BELA team conducted assessments of land condition and trends, evaluated ecosystem services for the interacting climate and policy scenarios, estimated the impact of ecosystem services on the macro economy, and identified landscape investment priority areas.

As part of the Global Program for Sustainability-funded Natural Capital Accounting initiative, the BELA team worked with the Ethiopian Ministry of Agriculture on developing an Investment Prioritization Tool (IPT) to measure ecosystem services, and identify priority watersheds for investment, based on the greatest potential to improve ecosystem services. A technical working group (TWG) was established to co-develop IPT and to build capacity for its use. The team also organized virtual training on the data and models underlying the IPT, and organized consultations at different stages of development of the tool to validate results with the government stakeholders.

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WHAT IS BELA?

The **Biodiversity, Ecosystems, and Landscape Assessment (BELA) initiative** supports landscape assessments of biodiversity and ecosystem services in World Bank engagements. BELA provides Bank teams and clients with efficient, tailored, and in-house analytical services, offering **insights on land condition & trends, landscape investment priority areas, ecosystem services in macro economy, climate change impacts, carbon market potential, and support for natural capital accounting**. The BELA team works on a contract basis with World Bank teams, with funding for core operations provided by the PROGREEN Global Partnership for Sustainable and Resilient Landscapes.

Interested in learning more? Click here.


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WHAT WAS THE IMPACT OF BELA'S WORK?


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BELA's analysis - landscape condition assessment and ecosystem services modeling, under different climate and policy scenarios - brought to light a notable shift in the hotspots of land degradation within the country. While the GoE's investments in the highlands have effectively reversed some concerning degradation trends, BELA's landscape assessments identified additional areas where focused adaptation efforts are still needed to address land degradation in the lowlands. This observation led to a key recommendation from the CCDR, namely, to increase investments in landscape management of strategic lowland areas. The IPT was developed to support this strategic shift, and is being used to identify priority watersheds for investment under the World Bank-financed Climate Action through Landscape Management (CALM) program. The tool will also be used to inform Ethiopia's Strategic Investment Framework for Sustainable Land Management that will inform government-financed programs on sustainable land management.

KEY QUESTIONS

- How will landscape conditions magnify or mitigate the impact of climate change on Ethiopia's economy and ecosystems?
- Where can investments in landscape management be prioritized to build climate resilience, alter the trajectory of land degradation, and provide positive economic benefits across sectors?
- What is the contribution of landscape degradation to Ethiopia's future carbon emissions and what is the potential for the country to participate in the global carbon market?

BELA APPROACH

Phase 1
PLANNING

- Meet with relevant stakeholders to align around common vision
- Define study objectives
- Identify and gather resources, data and experts

Phase 2
ECOSYSTEM SERVICES ANALYSIS

- Land condition & trends
- Landscape investment priority areas
- Ecosystem services in macro economy
- Climate change impacts
- Carbon market potential
- Support for natural capital accounting

Phase 3
RESULTS > INSIGHTS

- Land degradation and climate impacts
- Opportunity-based priority areas for investment
- Carbon offset and market potential

STAKEHOLDER ENGAGEMENT AND CAPACITY DEVELOPMENT

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Phase 1 PLANNING

BELA worked with the GoE's Ministry of Agriculture (MoA) to identify key ecosystem services and formulated diverse land management scenarios, considering varying degrees of resource pressures, urban expansion, and land management interventions. In parallel, BELA worked collaboratively with sectoral teams contributing to the CCDR. This collaboration aimed at establishing connections between land degradation, ecosystem services, and the macroeconomy, with a specific focus on addressing these aspects through the land management scenarios developed using the IPT.

To ensure synergy and coherence across these efforts, BELA coordinated activities among teams supporting Natural Capital Accounting (United Nations Statistics Division) under the Ethiopia Resilient and Green Development PASA (P178754), the Bank CCDR team, and Industrial Economics (IEc) consultants.

Data were compiled on soils, land cover, land condition, and management, climate, carbon, infrastructure, poverty, and costs of interventions from remote sensing sources, globally available data, from sectoral teams, and from local sources, when available.

Phase 2 ECOSYSTEM SERVICES ANALYSIS

The BELA team:

Conducted assessments of land condition and trends and developed a Business as Usual (BAU) scenario in which population pressures on resources continue and land condition reflects a continuation of past degradation trends.

Evaluated ecosystem services for the interacting climate and policy scenarios. Services for the CCDR analysis included erosion control and total land-based carbon storage and emissions offset by land management investments in each policy scenario. Services for the IPT included erosion control, water regulation, and carbon storage.

Estimated the impact of ecosystem services on the macro economy, by incorporating the erosion results into agriculture production models in collaboration with IEc.

Identified landscape investment priority areas, using ecosystem services models to target investments in sustainable land management and compared the impacts of targeted investment on agriculture production and **the potential for carbon finance**.

Phase 3 RESULTS --> INSIGHTS

BELA communicated its findings to both Bank staff and clients by integrating its outputs into the CCDR report. This integration not only enriched the report but also ensured that the critical insights derived from BELA's analysis were effectively disseminated.

Active participation in workshops and presentations further facilitated the dissemination of BELA's results. The team engaged in regular missions, collaborating closely with technical experts within the government and other stakeholders to co-develop the IPT. Recognizing the importance of knowledge transfer, BELA organized virtual training sessions focused on the underlying data and models integral to the IPT.

In a collaborative effort with the CCDR team, BELA participated in missions dedicated to validating results with government stakeholders. This hands-on involvement in the validation process underscored BELA's commitment to ensuring the accuracy and relevance of its findings, reinforcing the collaborative nature of its engagement with the government and other stakeholders.

BELA INSIGHTS

How will landscape conditions magnify or mitigate the impact of climate change on Ethiopia's economy and ecosystems?

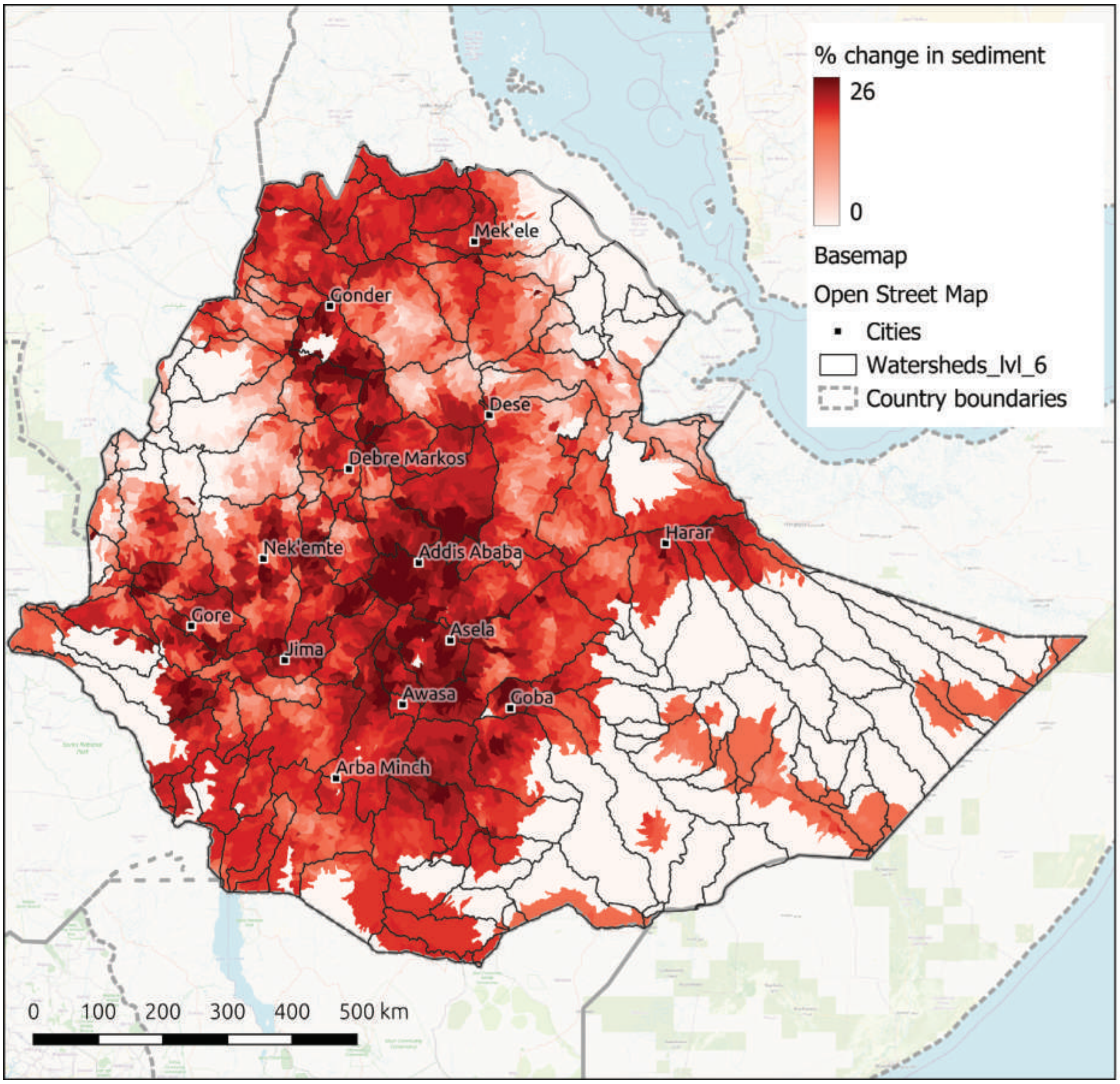

Figure 1. Spatial maps of percent change in sediment from current (2020) conditions to a projected Business as Usual future. (Source: BELA Initiative, World Bank).

In Ethiopia, half of the highlands are significantly degraded, impacting millions and weakening resilience to climate change. Yet, there has been little effort to quantify the effects of Sustainable Land Management (SLM) investments on ecosystem services or prioritize these investments strategically to optimize impacts. The CALM Program is part of the ongoing World Bank broad portfolio of support comprising around US$900 million for sustainable landscape management covering 8000 community watersheds in the country. The BELA analysis underscores the severity of the situation. Under a BAU scenario, erosion and sedimentation - responsible for land productivity decline and infrastructure threats - are projected to rise by more than 26 percent in numerous regions of Ethiopia (see Figure 1).

> Where can investments in landscape management be prioritized to build climate resilience, alter the trajectory of land degradation, and provide positive economic benefits across sectors?

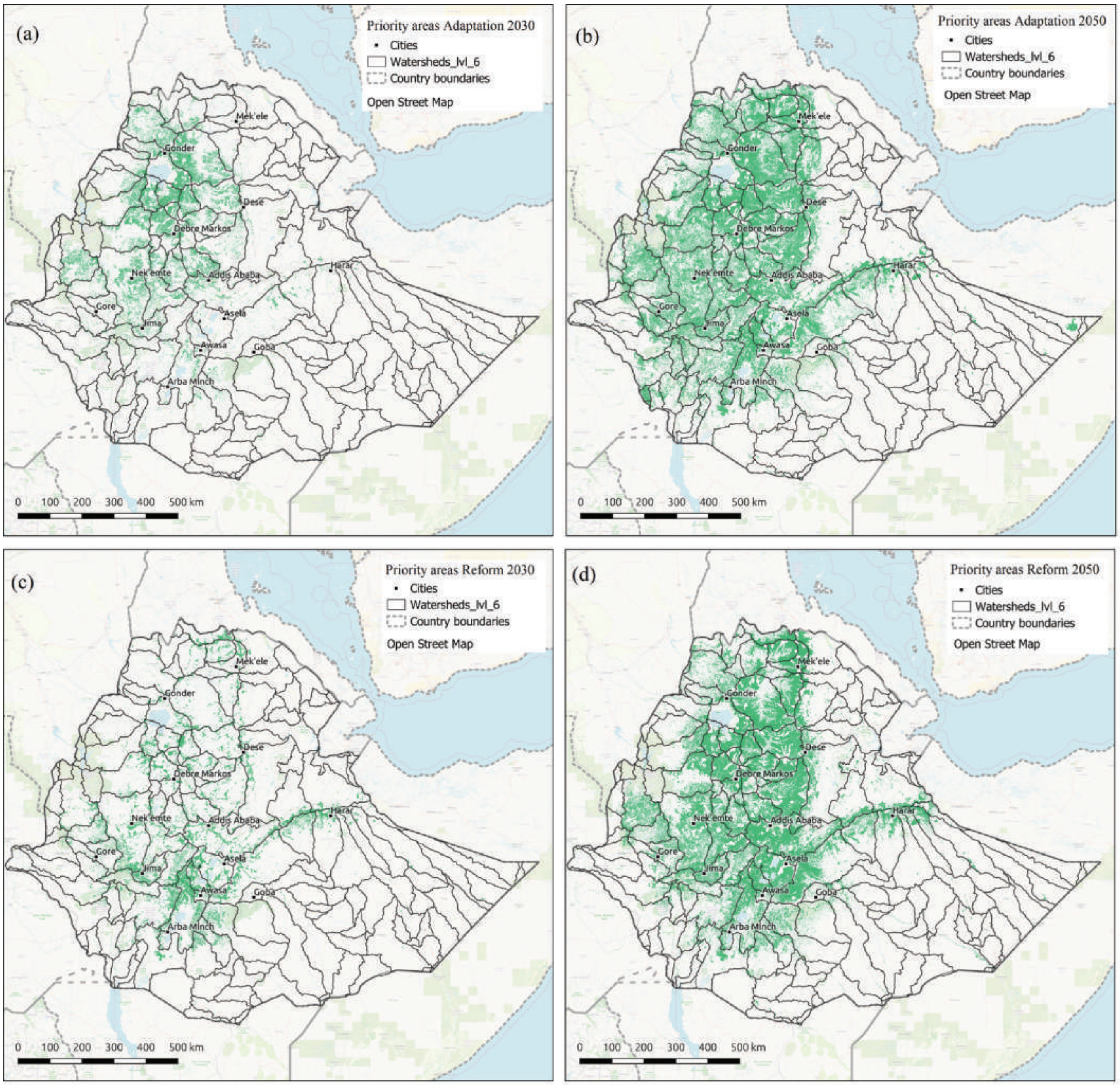

Figure 2. Spatial pattern maps of priority SLM intervention areas under the scenario of Adaptation 2030 (a) and 2050 (b), and Reform 2030 (c) and 2050 (d). (Source: BELA Initiative, World Bank).

Incorporating well-planned policies, regulations, and targeted investments in integrated land management is integral to the government's developmental blueprint, promising a substantial uplift in land conditions. The strategic insights provided by BELA's analysis have been visually mapped to identify priority areas under different future scenarios, as depicted in Figure 2. The Reform scenario (REF) envisions a forward-looking investment strategy in the agriculture sector, resulting in an annual 2 percent expansion of cropland and the extension of irrigation to cover 2.8 million hectares. Simultaneously, this scenario entails substantial investments in sustainable landscape management, aligning with the nation's ambitious restoration goals. On the other hand, the Adaptation scenario mirrors the REF scenario but takes a nuanced approach. Instead of solely targeting areas with high degradation, investments are strategically directed towards zones with the highest potential to enhance ecosystem services – specifically, erosion control, water regulation, and carbon sequestration. This approach aims to fortify landscape resilience in regions facing elevated climate change risks while maximizing benefits for rural populations.

Looking ahead to 2030 and 2050 (see Figures 2a and 2b respectively), the Adaptation scenario foresees the conversion of approximately 6.7 million hectares and 23.8 million hectares, inclusive of restoration and reforestation activities. This transformation spans 10 percent moist Afromontane, 30 percent Dry Afromontane, 5 percent Acacia, and 5 percent Combretum biomes. In contrast, the Reform scenario projects the conversion of about 6.7 million hectares and 23 million hectares during 2030 and 2050 (see Figures 2c and 2d respectively). This includes restoration and reforestation activities covering 20 percent moist Afromontane, 60 percent Dry Afromontane, 10 percent Acacia, and 10 percent Combretum. These scenarios outline a proactive vision for land management, reflecting a commitment to sustainable development and ecological resilience.

What is the contribution of landscape degradation to Ethiopia's future carbon emissions and what is the potential for the country to participate in the global carbon market?

The restoration of degraded landscapes emerges not only as a key developmental imperative but also as a potential avenue for mitigating greenhouse gas (GHG) emissions. This dual benefit could, in turn, generate resources capable of offsetting some of the expenses associated with land restoration efforts. According to BELA's findings, the Constrained Growth (CG) scenario - characterized by widespread land degradation - predicts a loss of 171 million metric tons of land-based CO_2 equivalent storage by 2030.

In contrast, initiatives aimed at enhancing landscape resilience in areas with the highest potential to improve ecosystem services (Resilient Scenario) paint a more promising picture. Here, carbon stocks are projected to increase by 389 million metric tons by 2050 – an increase 1.3 times greater compared to the Structural Reform Scenario and a substantial 6.7 times more than the CG scenario. The potential to sequester this additional carbon opens opportunities in the voluntary carbon markets, estimated to yield around US$130 million under the Resilient Scenario.

While this sum might appear modest in the face of the estimated US$21 billion required for comprehensive land management programs, it nonetheless represents a critical infusion of resources for developmental investments.

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FURTHER READING

- Ethiopia CCDR

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